

Vector Space

Internal Composition:

$\Rightarrow$ Let V be any set. Then the mapping $f: V \times V \rightarrow V$ is said to be internal composition and it is also called vector addition.

example:-

$$f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$
$$f(a, b) = a \cdot b$$

External Composition:

$\Rightarrow$ Let V and F be any two non-empty set. Then the mapping $f: V \times F \rightarrow V$ is said to be external composition in V over F also called scalar multiplication.

Vector Space:

$\Rightarrow$ Let $(F, +, \cdot)$ be a field. The elements of F will be called scalars. Let V be a non-empty set whose elements will be called vectors.

$\Rightarrow$ Then V is a vector space over the field F , if

i) There is defined an internal composition in V called addition of vectors and denoted by $+$.

$\Rightarrow$ Also for this composition V is an abelian group.

a) Closure property $\alpha + \beta \in V$ for all $\alpha, \beta \in V$

b) Commutative $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in V$

c) Identity $\exists$ an element $0 \in V$ such that $\alpha + 0 = \alpha$ for all $\alpha \in V$. This elem $0 \in V$ will be called the zero vector.

d) Associative $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$

e) Inverse To every vector $\alpha \in V$, there exist a vector $-\alpha \in V$ such that

$$\alpha + (-\alpha) = 0$$

ii) There is an external composition in V over F called scalar multiplication and denoted multiplicatively i.e. $a \alpha \in V$ for all $a \in F$ and for all $\alpha \in V$.

$\Rightarrow$ In other words V is called closed with respect to scalar multiplication

ii) The two compositions i.e. Scalars multiplication and addition of vectors satisfy the following postulates:

a) $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma \quad \forall \alpha \in F \text{ and } \beta, \gamma \in V$

b) $(\alpha + \beta)\gamma = \alpha\gamma + \beta\gamma \quad \forall \alpha, \beta \in F \text{ and } \gamma \in V$

c) $(\alpha\beta)\gamma = \alpha(\beta\gamma) \quad \forall \alpha, \beta \in F \text{ and } \gamma \in V$

d) $1 \cdot \alpha = \alpha \quad \forall \alpha \in V$ and 1 is unit element of the field F .

Q#:-

What is field?

Field:

A set F with two operations $+$, $\cdot$ is called a field if the following hold:

1) $\forall a, b \in F$

$a + b \in F$ closure property

2) $\forall a, b \in F$

$a + b = b + a$

3) $\forall a, b \in F$

$(a + b) + c = a + (b + c)$

4) $\forall a, b \in F$

$a + 0 = a$

5) $\forall a, b \in F$

$a \cdot 1 = a$

1)

$a \cdot b \in F$

$a \cdot b = b \cdot a$

$(a \cdot b) \cdot c = a \cdot (b \cdot c)$

$1 + 0 = 1$

$a \cdot \frac{1}{a} = 1$

In short: A set F which is abelian group under addition, abelian group under multiplication, and also satisfy the distributive property is called a field.
 eg - set of real no $F(\mathbb{R})$, set of rational no $F(\mathbb{Q})$, set of complex no $F(\mathbb{C})$

Q#: What is group?

$\Rightarrow$ A set G is called group if it satisfies the following

- o Closure property
- o Associative property
- o Identity property
- o Inverse property

Q#:- What is abelian group?

Abelian group:-

Let S be a non-empty set and $+$ be any binary operation on S , then the algebraic system $(S, +)$ is called the abelian group if it satisfies the following properties:

- i) Closure property
- ii) Associative property
- iii) Identity property
- iv) Inverse property
- v) Commutative property

Examples:

$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$ with $x = +$ and $a = 0$ are abelian groups.

Types of vector spaces:-

Real vector spaces:

$\Rightarrow$ Vector space with real scalars will be called real vector spaces.

Complex vector spaces:

$\Rightarrow$ Vector spaces with complex scalars are called complex vector spaces.

Zero vector spaces:

$\Rightarrow$ Let V consist of a single object which is denoted by 0 , and define $0+0=0$ and $k0=0$ for all scalars k . This is called zero vector space.

Note:

i) The set of all integers is not a vector space.

$1 \in V, 1/2 \in \mathbb{R}$: it is not closed

$(1/2)1 = 1/2 \notin V$ under scalar multiplication

Practice questions:

#1) Check whether the vector
 $V = \{x, y, z \in \mathbb{R}^3 : 2x + 3y^3 - 4z^2 = 0\}$ is
vector space or not?

(Given $V = \{x, y, z \in \mathbb{R}^3 : 2x + 3y^3 - 4z^2 = 0\}$)

So, take

$$(2, 0, 1), (8, 0, 2) \in V$$

$$= 2(2, 0, 1) + (8, 0, 2) = (10, 0, 3)$$
$$= -36 \neq 0$$

i.e. Adding vector $(10, 0, 3)$ does not
belong to V . Therefore, set V is not
a vector space.

#2) Check whether the set $V = \{(a, b, c) \in \mathbb{R}^3 : (a-b)c = 0\}$ is vector space or not?

(Given that $V = \{(a, b, c) \in \mathbb{R}^3 : (a-b)c = 0\}$)

So take, $(1, 1, 2), (2, 5, 0) \in V$

$$V_{S1} = (1, 1, 2) + (2, 5, 0) = (3, 6, 2)$$

$$\text{So, } (3-6)2 = -6 \neq 0$$

Therefore, closure property is not
satisfied. Thus, set V is not
vector space.

EX#: $V = \mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$,
 $F = \mathbb{R} = \text{set of real numbers}$

i) $(V, +)$ abelian group.

$$1) (x, y) + (a, b) = (x+a, y+b) \in V$$

$$3) (0, 0) + (x, y) = (x, y) \in V$$

$$4) (x, y) + (-x, -y) = (0, 0) \in V$$

2) $V \rightarrow (x, y) \in V, a, b \in F$
 $a_1(x, y) = a_1x, a_1y \in V$

3) $a, b \in V, a, b \in F$

$$a(a+b) = a[(x_1, y_1) + (x_2, y_2)]$$
$$= a[(x_1+x_2, y_1+y_2)]$$

$$= (ax_1+ax_2, ay_1+ay_2)$$
$$= (ax_1, ay_1) + (ax_2, ay_2)$$
$$= a(x_1, y_1) + a(x_2, y_2)$$
$$= ad + ab$$

4) $a, b \in F$ and $\alpha \in V$

$$= (a+b)\alpha$$

$$= a\alpha + b\alpha$$

$$= (a+b)(x_1, y_1)$$

$$= (a+b)x_1, (a+b)y_1$$

$$= (ax_1+bx_1, ay_1+by_1)$$

$$= a(x_1, y_1) + b(x_1, y_1)$$
$$= a\alpha + b\alpha$$

Q#:
Show that the set of all second degree polynomials is not a vector space.

Let

$$P(x) = -x$$

$$Q(x) = 2x + 1$$

$$\Rightarrow P(x) + Q(x) = -x + 2x + 1 = x + 1 \notin V$$

$\therefore$ it is not closed under scalar multiplication.

Define Subspace.

Subspace

$\Rightarrow$ A non-empty subset W of a vector space V over a field F is said to be a subspace of V if W is itself a vector space over F under the same operations as defined in V .

In other words:

$\rightarrow$ if W is a non-empty subset of a vector space V , then W is a subspace of V

if and only if the following conditions hold

- 1) if u and v are in W , then $u + v$ is in W .
- 2) if u is in W and c is any scalar, then cu is in W .

$\Rightarrow$ Every vector space has two subspaces

so V itself
Also called the set $\{0\}$ vector or null vector
trivial subspace.

Subspace Theorem

Let V be a vector space over a field F and let W be a non-empty subset of V . Then W is a subspace of V if and only if

- i) $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$ (closure property)
- ii) $w \in W, a \in F \Rightarrow aw \in W$ (scalar multiplication)

Problem #1: if V is a set of all $(n \times n)$ matrices over any field F , then show that W of all $(n \times n)$ symmetric matrices forms a vector subspace of $V(F)$.

Given:

$\Rightarrow$ Any matrix $A = [A_{ij}]_{n \times n}$ will be called a symmetric matrix, if $A = A^t$.

$\Rightarrow$ Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ be two symmetric matrices, then we have

$$\begin{aligned} A+B &= [a_{ij}]_{n \times n} + [b_{ij}]_{n \times n} \\ &= [a_{ij} + b_{ij}]_{n \times n} \\ &= [a_{ji} + b_{ji}]_{n \times n} \\ &= (A+B)^t \end{aligned}$$

ii) if $a \in F$ and $A = [a_{ij}]_{n \times n} \in W$
Now,

$$\begin{aligned} a \cdot A &= a \cdot [a_{ij}]_{n \times n} \\ &= [a \cdot a_{ij}]_{n \times n} \\ &= [a \cdot a_{ji}]_{n \times n} \\ &= (a \cdot A)^t \end{aligned}$$

$a \cdot A$ is a symmetric matrix.

$\Rightarrow$ Hence, W is closed under scalar multiplication & addition.

$\Rightarrow$ Hence $W(F)$ is a vector subspace of $V(F)$.

Problem #2: Show that the set $W = \{ (a, b, 0) : a, b \in F \}$ is a subspace of $V_3(F)$.

Let $\alpha, \beta \in W, c \in F$

Then

$$\alpha = (a_1, b_1, 0)$$

$$\beta = (a_2, b_2, 0)$$

for some $a, b \in F$

$$\alpha + \beta = (a_1 + a_2, b_1 + b_2, 0) \in W$$

$$c\alpha = (ca_1, cb_1, 0) \in W$$

Hence W is a subspace of $V_3(F)$.

Q# Any subset of a set of polynomials is also space.

Let P and v be the polynomials

$$P(x) = a_0 + a_1x + \dots + a_nx^n$$

and

$$v(x) = b_0 + b_1x + \dots + b_nx^n$$

$$i) (P+v)x = P(x) + v(x)$$

$$= (a_0+b_0) + (a_1+b_1)x + \dots + (a_n+b_n)x^n$$

$\in W$

$$ii) kP(x) = k a_0 + k b_1x + \dots + k b_nx^n$$

$\in W$

- Note: Collection of all skew matrices forms a vector subspace of $V(F)$.

Note: Collection of all orthogonal matrices do not form a vector space of $V(F)$ because zero vector is not in this set.

Note: $\{A \mid \det(A) = 0\}$ forms a subspace of $V(F)$

$$\Rightarrow \vec{0} \in W$$

$$\det(\vec{0}) = 0$$

$$\text{Let } A, B \in W, \alpha \in \mathbb{R}$$

$$i) \alpha A + B \in W$$

$$\det(\alpha A + B) = \det(\alpha A) + \det(B)$$

$-\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$

$$\alpha \det(A) + \det(B) = \alpha \det(0) + \det(0) = 0$$

Note: $\det(A) \neq 0$ do not form a subspace. bcz $\vec{0} \notin W$

$$\det(\vec{0}) = 0$$

Note: $|A| = 0$ and $|A| \neq 0$ do not form a subspace

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|A| = 0, |B| = 0$$

$$A+B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \neq 0$$

Q# Show that union of two subspaces is not necessarily a subspace.

Let the vector space $\{(x, y, z) \mid x, y, z \in \mathbb{R}\}$

$$W_1 = \{(x, 0, 0) \mid x \in \mathbb{R}\}$$

$$W_2 = \{(0, y, 0) \mid y \in \mathbb{R}\}$$

$$\text{Let } U = (x_1, 0, 0) \in W_1$$

$$\text{and } V = (0, x_2, 0) \in W_2, x_1, x_2 \in \mathbb{R}$$

$$\text{Let } \alpha, \beta \in \mathbb{R} \text{ and } U, V \in W_1$$

$$\alpha U + \beta V = \alpha(x_1, 0, 0) + \beta(0, x_2, 0)$$

$$= (\alpha x_1, 0, 0) + (0, \beta x_2, 0)$$

$$= (\alpha x_1 + \beta x_2, 0, 0) \in W_1$$

$\Rightarrow$ for all $\alpha, \beta, x_1, x_2 \in \mathbb{R} \Rightarrow \alpha x_1 + \beta x_2 \in W_1$

$\Rightarrow W_1$ is subspace of $V_3(\mathbb{R})$.

$\Rightarrow$ Hence W_1 is also subspace of $V_3(\mathbb{R})$.

Now we show that $W_1 \cup W_2$ is not subspace.

Let $\alpha, \beta \in \mathbb{R}$ and

$$W_1 = (x, 0, 0) \in W_1$$

$$W_2 = (0, y, 0) \in W_2$$

$$\Rightarrow W_1 = (x, 0, 0) \in W_1 \cup W_2$$

$$\text{and } W_2 = (0, y, 0) \in W_1 \cup W_2$$

$$\left[\begin{array}{l} W_1 \subseteq W_1 \cup W_2 \\ W_2 \subseteq W_1 \cup W_2 \end{array} \right]$$

$$\alpha W_1 + \beta W_2 = \alpha(x, 0, 0) + \beta(0, y, 0)$$

$$= (\alpha x, 0, 0) + (0, \beta y, 0)$$

$$\alpha W_1 + \beta W_2 = (\alpha x, \beta y, 0)$$

Now $\alpha W_1 + \beta W_2 \notin W_1$ as well as

$$\alpha W_1 + \beta W_2 \notin W_2$$

$$\Rightarrow \alpha W_1 + \beta W_2 \notin W_1 \cup W_2$$

Hence $W_1 \cup W_2$ is not subspace.

e.g

$$\alpha = (1, 2, 0)$$

$$\beta = (3, 0, 5)$$

for two scalars $a=1, b=2$

Show that Union of two subspaces is a subspace iff and only if one is contained in other.

Let W_1 and W_2 are two subspaces.

we suppose that one subspace is contained in other

i) when W_1 contained in W_2

$$W_1 \subseteq W_2 \Rightarrow W_1 \cup W_2 = W_2$$

ii) when W_2 contained in W_1

$$W_2 \subseteq W_1 \Rightarrow W_1 \cup W_2 = W_1$$

$\Rightarrow$ in both cases $W_1 \cup W_2$ is subspace.

Converse, we suppose that

" $W_1 \cup W_2$ " is a subspace and neither W_1 contained in W_2 nor W_2 contained in W_1 .

i.e an element x such as

$$x \in W_1 \text{ but } x \notin W_2$$

and

$$y \in W_2 \text{ but } y \notin W_1$$

$$\Rightarrow x, y \in W_1 \cup W_2 \Rightarrow x+y \in W_1 \cup W_2$$

$$\Rightarrow (x+y) \in W_1 \text{ or } (x+y) \in W_2 \text{ or both}$$

i) if $(x+y) \in W_1$, $-x \in W_1$
 $\Rightarrow (x+y) - x \in W_1 \Rightarrow 0+y \in W_1 \Rightarrow y \in W_1$

This is contradicted result of our assumption.

ii) if $(x+y) \in W_2$, $-y \in W_2$
 $(x+y) - y \in W_2 \Rightarrow x \in W_2$

it is also contradicted result of our assumption

i.e. our assumption is wrong.

Hence, opposite of our assumption is true which is either W_1 contained in W_2 or W_2 contained in W_1 .

Q.H Show that intersection of two subspaces is also a subspace.

Let 'F' be a field and 'V' is a vector space over 'F' $V(F)$

Also, consider 'W₁' and 'W₂' are two subspaces of 'V'

i.e. if additive identity is

0 then '0' $\in W_1$ and '0' $\in W_2$

$\Rightarrow$ '0' $\in (W_1 \cap W_2)$

Hence intersection of two subspaces $(W_1 \cap W_2)$ is non-empty subset of 'V'.

Let $x, y \in (W_1 \cap W_2)$ and $\alpha, \beta \in F$

$\Rightarrow x, y \in W_1$ and $x, y \in W_2$

$\Rightarrow \alpha x + \beta y \in W_1$ and $\alpha x + \beta y \in W_2$

$\Rightarrow \alpha x + \beta y \in W_1 \cap W_2$

$\forall x, y \in (W_1 \cap W_2)$ and $\alpha, \beta \in F$

Hence, $W_1 \cap W_2$ is also a subspace from theorem 2.

Q.H: Show that any subset of an infinite space is also a space.

$\Rightarrow$ By fundamental theorem of calculus the sum of continuous function is continuous and that of constant time.

$\Rightarrow$ if any continuous function is repeated it is also continuous function.

$\Rightarrow$ Hence, the set of all continuous function $(-\infty, +\infty)$ is a subspace of the function $f: (-\infty, +\infty) \rightarrow \mathbb{R}$.

Note: Set of invertible matrices does not make a subspace as it does not contain

- o it is not closed under addition
- o it is not closed under multiplication

Q: What is Column space?

Column space:

- $\Rightarrow$ The column space consists of all linear combinations of the columns.
- $\Rightarrow$ The combinations are all possible vectors Ax . They fill the column space.
- $\Rightarrow$ if A is $m \times n$ and $A = [a_1, a_2, \dots, a_n]$
 $\text{Col}(A) = \text{Span}\{a_1, a_2, \dots, a_n\}$
 $= \{b \mid b = Ax \text{ for some } x \in \mathbb{R}^n\}$

Example: Find a matrix A such that $w = \text{Col}(A)$

$$w = \left\{ \begin{bmatrix} -a+2b \\ a-2b \\ 3a-6b \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

$$= \text{Span}(A)$$

$$= \left\{ a \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix} + b \begin{bmatrix} 2 \\ -2 \\ -6 \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

$$= S(v_1) + S(v_2)$$

$$A = [v_1 \ v_2] = \begin{bmatrix} -1 & 2 \\ 1 & -2 \\ 3 & -6 \end{bmatrix}$$

Q11: Find A so that $\text{Col}(A) = w$.
 if $\text{Col}(A)$ is a subspace of $\mathbb{R}^4$
 what is k ?

$$w = \left\{ \begin{bmatrix} b-c \\ ab+cd \\ 5c-4d \\ d \end{bmatrix} \right\}$$

$$w = \left\{ b \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ 5 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \\ -4 \\ 1 \end{bmatrix} : b, c, d \in \mathbb{R} \right\}$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix} \quad m \times n \quad \mathbb{R}^m$$

$= 4 \times 3 \quad \mathbb{R}^4$

Note:

- o A subspace of a vector space V is a subset H that satisfies the conditions:
 - o The zero vector 0 in V
 - o $0 \in H$
 - o H is closed under addition: $u, v \in H \Rightarrow u+v \in H$
 - o H is closed under multiplication of scalars: $c \in \mathbb{R}, u \in H \Rightarrow cu \in H$

What is NULL Space?

NULL Space

$\Rightarrow$ The solution space of the homogeneous system of equations $AX=0$, which is a subspace of R^n , is called the null space of A .

$$NUL(A) = \{x : x \in R^n \text{ and } Ax=0\}$$

$\Rightarrow$ Elementary row operation do not change null space.

What is row Space?

Row Space

$\Rightarrow$ if A is $m \times n$ matrix, then the subspace of R^n spanned by the row vectors of A is called the row space of A , and the subspace of R^m spanned by the column vectors of A is called the column space of A .

$\Rightarrow$ Elementary row operation do not change the row space of a matrix.

$\Rightarrow$ For an $n \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix}$$

The vectors

$$x_1 = [a_{11} \ a_{12} \ \dots \ a_{1n}]$$

$$x_2 = [a_{21} \ a_{22} \ \dots \ a_{2n}]$$

$$\vdots$$

$$x_m = [a_{m1} \ a_{m2} \ \dots \ a_{mn}]$$

R^n that are formed from the rows of A are called the row vectors of A .

$$C_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, C_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, C_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

R^m is formed from the columns of A are called the column vectors of A .

Example:

Find row and column vectors of 2×3 matrix.

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & -1 & 4 \end{bmatrix}$$

$$x_1 = [2 \ 1 \ 0] \text{ and } x_2 = [3 \ -1 \ 4]$$

$$C_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, C_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, C_3 = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

Q#: Is $U = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$ is NUL(A),

where $A = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}$?

$$AU = 0$$

$$= \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$$

$$= 5 \begin{bmatrix} 5 \\ 13 \\ 8 \end{bmatrix} + (-3) \begin{bmatrix} 21 \\ 23 \\ 14 \end{bmatrix} + 2 \begin{bmatrix} 19 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 25 \\ 65 \\ 40 \end{bmatrix} + \begin{bmatrix} -63 \\ -69 \\ -42 \end{bmatrix} + \begin{bmatrix} 38 \\ 4 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \underline{0}$$

Topic #:-

Matrix Transformation for

Standard basis:

Definition- if T is a function with domain R^n and codomain R^m , then we say that T is a transformation from R^n to R^m or that T maps from R^n to R^m , which is denoted by:

$$T: R^n \rightarrow R^m$$

a transformation is sometimes called an operator on R^n .

$\Rightarrow$ Matrix transformation is denoted by:

$$T_A: R^n \rightarrow R^m$$

Theorem -

$T: R^n \rightarrow R^m$ is a matrix transformation if and only if the following relationship hold for all vectors U and V in R^n and for every scalar k :

- $T(U+V) = T(U) + T(V)$ (Additive property)
- $T(kU) = kT(U)$ (Homogeneity property)

Theorem:-

Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a matrix transformation, and every matrix transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a linear transformation.

Examples #:

$$T(x, y, z) = (2x - 4y + 9z, 5x + 3y - 2z)$$

for $\mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$e_1 = (1, 0, 0)$$

$$\begin{aligned} T(e_1) &= (2(1) - 4(0) + 9(0), 5(1) + 3(0) - 2(0)) \\ &= (2, 5) \end{aligned}$$

$$e_2 = (0, 1, 0)$$

$$\begin{aligned} T(e_2) &= (2(0) - 4(1) + 9(0), 5(0) + 3(1) - 2(0)) \\ &= (-4, 3) \end{aligned}$$

$$e_3 = (0, 0, 1)$$

$$\begin{aligned} T(e_3) &= (2(0) - 4(0) + 9(1), 5(0) + 3(0) - 2(1)) \\ &= (9, -2) \end{aligned}$$

$$A = [T(e_1) \ T(e_2) \ T(e_3)] = \begin{bmatrix} 2 & -4 & 9 \\ 5 & 3 & -2 \end{bmatrix}$$

Q#:-

Find the standardized matrix A for the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + x_2 \\ x_1 - 3x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2(1) + 0 \\ 1 - 3(0) \\ -1 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2(0) + 1 \\ 0 - 3(1) \\ -0 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix}$$

$$A = [T(e_1) | T(e_2)] = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ -1 & 1 \end{bmatrix}$$

Q#:-

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x_1, x_2, x_3) = (x_1, x_2 - x_3)$$

$$e_3 = (0, 0, 1)$$

$$T(e_3) = (0, -1)$$

$$e_2 = (0, 1, 0)$$

$$T(e_2) = (0, 1)$$

$$e_1 = (1, 0, 0)$$

$$T(e_1) = (1, 0)$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$